OpenADMET PXR Challenge — Model Description Report

Submission: Phase 2 blind test set (260 compounds) + Phase 1 test set (253 compounds)
Total: 513 compounds
Predicted property: PXR pEC50 ($-\log_{10}$ EC_{50}, in molar units)

Training Data

The model was trained on a combined dataset of **4,827 compounds** assembled from three sources:

1. **OpenADMET competition training data (Phase 1):** The full labelled set provided by the challenge organisers, comprising proprietary PXR pEC50 measurements at physiologically relevant assay conditions.
2. **HTChem counter-screen compounds:** Crude and semi-purified compounds from the HTChem screening library were included. For crude samples, pEC50 values were yield-corrected to account for compound impurity prior to inclusion in the training set.
3. **ChEMBL PXR agonist data:** 508 compounds with PXR agonist activity were retrieved from ChEMBL and added to augment chemical space coverage. Only assay records obtained using a technology directly comparable to the competition assay (cell-based, transactivation, fluorescence-based reporter) were retained. Assays with divergent readouts or endpoints were excluded to maintain data consistency.

Structure Preparation

All training and test structures were prepared using **Schrödinger LigPrep**, generating low-energy 3D conformers with protonation states assigned at **pH 7.4**. LigPrep outputs were subsequently reviewed for systematic errors, and four categories of incorrect protonation were corrected by custom RDKit scripts applied sequentially:

Script	Error corrected
`neutralize_sulfonamide_N.py`	Sulfonamide N^- (R-SO_2-[N^-]-R) → neutral NH
`fix_phenol.py`	Deprotonated phenol (c-[O^-]) → cOH
`fix_amide.py`	Inverted amide (N=C[O^-]) → NC(=O) with corrected bond orders
`neutralize_nplus.py`	Pyridinium-type aromatic N^+H → neutral aromatic N

Quaternary N^+ centres (permanently charged, no hydrogen) were left unchanged. Identical fix scripts were applied to both training and test structures to ensure consistency.

Model Architecture and Training

A **ChemProp message-passing neural network (MPNN)** was used, augmented with RDKit 2D physicochemical descriptors concatenated to the graph-level representation before the feedforward network (FFN). Descriptors with zero variance or non-finite values across the training set were removed prior to training; the retained descriptor vector was standardised (zero mean, unit variance).

Hyperparameter optimisation was performed by 5-fold stratified cross-validation (stratified on pEC50 quintiles) over a grid spanning FFN hidden dimension {300, 512, 1024}, FFN depth {2, 3}, dropout {0.0, 0.2}, message-passing depth {3, 4}, and MP hidden dimension {300, 600, 1024}. Early stopping (patience = 10 epochs, max 50) was applied per fold.

Counter-assay weighting: Training compounds for which a counter-screen pEC50 was available were reweighted to emphasise selectivity for PXR. Compounds with a high PXR/counter-assay activity differential received a higher sample weight (up to 2.0); compounds with low selectivity received a down-weight (minimum 0.5). Compounds lacking counter-screen data were assigned a neutral weight of 1.0.

Ensemble: The best hyperparameter configuration was used to train **8 independent models** with different random seeds (42, 123, 456, 789, 1337, 2024, 31415, 99999) on the full training set. Learning rate followed a warmup schedule (initial 1×10^{-4}, peak 2×10^{-4}, final 1×10^{-5}). Final predictions are the mean of the 8-member ensemble; the standard deviation across members is reported as a calibrated uncertainty estimate.

Submitted Predictions

Phase 1 test compounds (253 compounds) are submitted with their experimentally measured pEC50 values. Phase 2 blind test compounds (260 compounds) are submitted with ensemble-mean predicted pEC50 values from the ChemProp model described above.